

Writing Large Numbers in Scientific Notation

Teacher's Edition

Read This First

This lesson rebuilds Math Nation 3.8 (SE pp. 139–145) and opens MA.8.NSO.1.4. The book's sequence — powers of ten and place value, the definition, the coefficient constraint, then 'how many times larger' in three cases — is kept as Notes I–III. Its Florida funding, painting, alligator and 8.4×10^6 items are kept as the examples and whiteboard 7.

Two things the state guide expects that the book does not ask are added: calculator E notation (whiteboard 8, bank 7c) and the ' 10^3 times is not 3 times' misconception as a diagnostic (whiteboard 6). The guide's own items — the blue whale and elephant, Florida and Lake Okeechobee, China and Mexico — are bank 9 and Additional 9.

Two book items were not reused. The Additional Practice key gives $10^6 + 10^4 = 101000$ (it is 1,010,000). Homework 2 scaffolds $8.1 \div 2.7 = 3$ and $10^8 \div 10^6 = 100$ and then asks about 8.1×10^8 versus $5.4 \times 10^7 = 15$, so the scaffold does not lead to the question; bank 8 uses a scaffold that does.

Coefficient constraint: every scientific-notation coefficient in this lesson is in $[1, 10)$ except where an item is ABOUT the constraint (whiteboards 1, 3, 5, 9; bank 11–13). Those items are tagged in the spec and were checked by hand.

The book's warm-up is the grains-of-sand story — good motivation, no mathematics until question 2. The number (7.5×10^{18}) is used in Notes I and as Additional 7a; the warm-up itself is spaced retrieval.

Standard and Targets

MA.8.NSO.1.4 Express numbers in scientific notation to represent and approximate very large or very small quantities. Determine how many times larger or smaller one number is compared to a second number.

Benchmark Express numbers in scientific notation to represent and approximate very large or very small quantities. Determine how many times larger or smaller one number is compared to a second number.

Boundary Large numbers today; small numbers (negative exponents) tomorrow in 3.09. Operations in scientific notation are MA.8.NSO.1.5, Unit 4.

Learning target I can write a large number in scientific notation, read one back into standard form, and find how many times larger one number is than another.

Essential question How does a power of 10 let us write a very large number briefly, and compare two of them?

Building on MA.7.NSO.1.1 and MA.8.NSO.1.3 (3.01–3.07) — powers of 10 and the laws; Grade 5 — place value to the billions.

Working toward 3.09 — small numbers and negative powers of ten; MA.8.NSO.1.5 (Unit 4) — operations in scientific notation.

Where this sits in the course First of two MA.8.NSO.1.4 lessons (3.08–3.09). The Grade 8 FAST reports this under Number Sense and Operations. The guide's named misconceptions — coefficient outside $[1, 10)$, 'E' read as an error, ' 10^4 times' read as '4 times', and multiplying the exponent when scaling — are each a distractor somewhere in today's round or bank.

Lesson at a Glance

Windy Hill periods run **53 minutes**. The whiteboard round takes the remainder after the taught blocks — **19 minutes** in this lesson — and IXL takes the last 5. Slide numbers below are read from the deck when this edition is built.

Segment	Slides	Time
Settle in; post the learning target	1	1 min
Warm-Up	2–3	5 min
Notes	4–6	11 min
Example 1	7–8	3 min
Your Turn	9–10	3 min
Example 2	11–12	3 min
Your Turn	13–14	3 min
Whiteboards	15–32	19 min
IXL	33	5 min
Total		53 min

Materials Whiteboards and markers. Calculators are allowed on everything; today is the day to show the class what E means on the display — enter 3.5×10^9 and 35,000,000,000 and look.

Where the lesson lives Notes II — 'one nonzero digit in front of the decimal point'. Whiteboard 5 is the diagnostic; whiteboard 9 asks for it in writing. Notes III's ' 10^3 times is a thousand times' is the second line to say out loud; whiteboard 6 tests it.

No partner or group work anywhere in this lesson. Every Math Nation exploration this lesson draws on was written for pairs; here the same tables are worked from the front and on individual boards.

Vocabulary

scientific notation — a number written as a product $a \times 10^n$ where a is at least 1 and less than 10, and n is an integer: 2.6×10^7

standard form — the number written out with all of its digits: 26,000,000

coefficient — the factor a in $a \times 10^n$; it must satisfy $1 \leq a < 10$

Warm-Up — spaced retrieval

Four questions, two minutes silent, then the reveal. No reteach longer than one sentence per question.

1. Rewrite with a positive exponent: $4x^{-3}$ **$4/x^3$**

retrieves: yesterday — T-A2 — only the factor with the negative exponent moves

2. Find the value: $(10^2)^3$ **$10^6 = 1,000,000$**

retrieves: last week — 3.03 — power of a power with base 10

3. Find the value: $\sqrt{121}$ **11**

retrieves: last unit — Unit 2 — perfect squares (2.01)

4. In 4,500,000, which place is the 4 in? **millions**

retrieves: prior grade — the prerequisite this lesson needs — Grade 5 — place value (MA.5.NSO.1): the 4 is in the millions place, the 5 in the hundred thousands

Slide-by-Slide Teaching Notes

Slide 1. Writing Large Numbers in Scientific Notation 1 min

- Post the learning target. Say the 'today' line and nothing else yet.

Slide 2. Warm-Up — On your own. Four minutes. No notes. 4 min

- Q2 and Q4 are the two halves of today: a power of 10 names a place. 10^6 is a million, and the 4 in 4,500,000 sits in the millions place, so 4,500,000 is 4.5 millions — 4.5×10^6 . Say that sentence once at the reveal and the lesson is half taught.

Slide 3. Warm-Up — Answers. 1 min

- Reveal. Ask for the band each question came from only if time allows.

Slide 4. Notes — Copy the table. 3 min

- The exponent is the number of zeros, and it is also the place of the 1: 10^3 is one thousand, and the 1 sits in the thousands place. Students fill the last two cells. Then the book's warm-up number: seven quintillion, five hundred quadrillion grains of sand on Earth's beaches, 7,500,000,000,000,000 — twenty digits, and nobody wants to copy it twice.

Slide 5. Notes 4 min

- Definition first, then the two examples. 2,000,000 is 2 millions: 2×10^6 . 26,000,000 is 2.6 ten-millions: the 2 is in the ten-millions place, so 2.6×10^7 . The rule of thumb — count how many places the decimal point moves — is on the last line, and it agrees with place value every time. Then the constraint, which the state guide names as the first misconception: 20×10^5 equals 2,000,000 but is NOT scientific notation, because 20 is not less than 10.
- **OFF THE SLIDE, YOURS TO SAY:** 'One nonzero digit in front of the decimal point. That is the whole rule.'

Slide 6. Notes 4 min

- Three cases, one line each. Same power of 10: divide the coefficients. Same coefficient: divide the powers — and 10^3 times is one thousand times, not three

times, which is the guide's second named misconception. Both different: divide both and multiply the results. The last line is the calculator: $8.4\text{E}6$ means 8.4×10^6 ; E is not an error.

- **OFF THE SLIDE, YOURS TO SAY:** ' 10^3 times larger is a thousand times larger. The exponent is not the answer; the power is.'

Slide 7. Example 1 — Standard form to scientific notation. 1 min

- SPOKEN ONLY. The book's Florida education-funding figure. Ask which digit goes in front of the decimal point and which place it is in before anyone writes.

Slide 8. Example 1 — Worked. 2 min

- The 3 is in the ten-billions place: 10^{10} . The coefficient keeps every nonzero digit after it — 3.01, not 3.1 and not 3. A board with 3.1×10^{10} dropped the zero between the 3 and the 1; a board with 3.01×10^9 counted the zeros after the 1 instead of the places the point moved.

Slide 9. Your Turn — Same steps, your numbers. Boards up when done. 2 min

- The book's auction painting. Catches 1.15×10^8 (dropped the zero) and 1.105×10^7 (counted the zeros after the 5). 1.105×10^8 is right.

Slide 10. Your Turn — Answer. 1 min

- Reveal; name what a wrong board most likely did (see the note above).

Slide 11. Example 2 — How many times larger? 1 min

- SPOKEN ONLY. Coefficients differ and powers differ. Ask for the plan — divide both, multiply the results — before anyone touches a calculator.

Slide 12. Example 2 — Worked. 2 min

- $8.1 \div 5.4 = 1.5$ and $10^8 \div 10^7 = 10$. So 15 times. A board with 1.5 stopped at the coefficients; a board with 10 stopped at the powers; a board with 150 wrote 10^2 for $10^8 \div 10^7$. Confirm in standard form: $810,000,000 \div 54,000,000 = 15$.

Slide 13. Your Turn — Same steps, your numbers. Boards up when done. 2 min

- The book's alligators. $1.5 \div 2.0 = 0.75$ and $10^6 \div 10^5 = 10$, so 7.5 — a case where the coefficient ratio is less than 1 and the answer is still 'times larger'. A board with 0.75 forgot the powers.

Slide 14. Your Turn — Answer. 1 min

- Reveal; name what a wrong board most likely did (see the note above).

Slide 15. Whiteboards · Question 1 of 9 19 min for the round

- One digit, six zeros. A board with 4×10^7 counted seven digits instead of six zeros; a board with 40×10^5 broke the coefficient rule.

Slide 16. Whiteboards · Question 1 of 9

- Reveal. Scan the back row first; question the blank boards before the wrong ones.

Slide 17. Whiteboards · Question 2 of 9

- Reading the notation back. A board with 720,000 moved five places (counted the 2 as a zero); a board with 7.20000 attached zeros without moving the point.

Slide 18. Whiteboards · Question 2 of 9

- Reveal. Scan the back row first; question the blank boards before the wrong ones.

Slide 19. Whiteboards · Question 3 of 9

- Three nonzero digits. 356×10^3 is equal in value but not scientific notation; 3.56×10^3 counted only the zeros.

Slide 20. Whiteboards · Question 3 of 9

- Reveal. Scan the back row first; question the blank boards before the wrong ones.

Slide 21. Whiteboards · Question 4 of 9

- The coefficient is a trap: 9.5 is bigger than 1.2, but 10^6 beats 10^5 . Compare the powers first, the coefficients only when the powers tie.

Slide 22. Whiteboards · Question 4 of 9

- Reveal. Scan the back row first; question the blank boards before the wrong ones.

Slide 23. Whiteboards · Question 5 of 9

- Diagnostic. B is the guide's first misconception and the one to name; C and D are the two counting errors.

Slide 24. Whiteboards · Question 5 of 9

- Reveal. Scan the back row first; question the blank boards before the wrong ones.

Slide 25. Whiteboards · Question 6 of 9

- Diagnostic — the guide's misconception in its purest form. A board full of B means Notes III's last line did not land: 10^3 is a thousand.

Slide 26. Whiteboards · Question 6 of 9

- Reveal. Scan the back row first; question the blank boards before the wrong ones.

Slide 27. Whiteboards · Question 7 of 9

- The book's own comparison. A board with 4 stopped at the coefficients; a board with 40 wrote 10 for 10^2 . Both are half an answer.

Slide 28. Whiteboards · Question 7 of 9

- Reveal. Scan the back row first; question the blank boards before the wrong ones.

Slide 29. Whiteboards · Question 8 of 9

- The state guide's item shape that the book never asks. E9 means $\times 10^9$. A board with 3.5^9 read E as an exponent on 3.5; a board with 35,000,000,000 moved the point ten places.

Slide 30. Whiteboards · Question 8 of 9

- Reveal. Scan the back row first; question the blank boards before the wrong ones.

Slide 31. Whiteboards · Question 9 of 9

- Written. The book's Guided Instruction item 3, kept. Full credit: both answers with the reason (20 is not less than 10) and the corrected form 2×10^6 . A student who says 'it is wrong' has not separated value from form — that separation is the point of the item.

Slide 32. Whiteboards · Question 9 of 9

- Reveal. Scan the back row first; question the blank boards before the wrong ones.

Slide 33. IXL — Last five minutes. 5 min

- IXL: whatever is not finished is tonight's practice; SmartScore 67.

The Whiteboard Round — what each board tells you

Nine questions, one at a time, each with its reveal. Boards down until the cue, all up together, scan the back row first, question the blank boards before the wrong ones. Question 9 is written work. Every wrong option on a multiple-choice question is a named error; the benchmark it comes from is cited.

#	Question	Answer	What a wrong board says
1	Write 4,000,000 in scientific notation.	4×10^6	4×10^7 — counted the digits instead of the places the point moves [B1G-M 8.NSO.1.4: exponent = places moved]; 40×10^5 — coefficient not in [1, 10] [B1G-M 8.NSO.1.4 misconception: coefficient outside 1 to 10]
2	Write 7.2×10^4 in standard form.	72,000	720,000 — the point was moved five places [8.NSO.1.4: 10^4 moves the point four places]; 7.20000 — zeros appended without moving the point
3	Write 356,000 in scientific notation.	3.56×10^5	356×10^3 — coefficient not in [1, 10] [B1G-M 8.NSO.1.4 misconception]; 3.56×10^3 — counted the zeros instead of the places the point moves
4	Which is greater, 9.5×10^5 or 1.2×10^6 ?	1.2×10^6	9.5×10^5 — compared the coefficients and ignored the powers [B1G-M 8.NSO.1.4: order of magnitude first]
5	Which is 52,000,000 written in scientific notation?	$A - 5.2 \times 10^7$	(B) equal in value, but the coefficient 52 is not less than 10 [B1G-M 8.NSO.1.4 misconception: coefficient outside 1 to 10]; (C) counted the six zeros instead of the seven places the point moves [8.NSO.1.4: exponent = places moved]; (D) counted the eight digits [8.NSO.1.4: exponent = places moved]

#	Question	Answer	What a wrong board says
6	How many times larger is 6×10^8 than 6×10^5 ?	A — 1,000	(B) 10^3 times read as 3 times [B1G-M 8.NSO.1.4 misconception: ' 10^4 times' read as '4 times']; (C) 3×100 — the exponent difference multiplied by a stray 100 [8.NSO.1.4: the ratio of the powers is 10^3]; (D) $8 \div 5$ — the exponents were divided [8.NSO.1.3: quotient of powers subtracts the exponents]
7	How many times larger is 8.4×10^6 than 2.1×10^4 ?	400	4 — coefficients only [8.NSO.1.4: both factors are compared]; 40 — 10^2 read as 10 [B1G-M 8.NSO.1.4 misconception: exponent read as the multiplier]
8	A computer displays a number as 3.5E9. Write the number in standard form.	3,500,000,000	'error' or 3.5^9 — E read as something other than $\times 10^9$ [B1G-M 8.NSO.1.4 misconception: calculator E notation]; 35,000,000,000 — point moved ten places
9	Marcus writes 2,000,000 as 20×10^5 . Is his expression equal to 2,000,000? Is it scientific notation? Explain.	Equal, yes. Scientific notation, no: 20 is not less than $10. 2 \times 10^6$.	'not equal' — value confused with form [8.NSO.1.4: equivalent expressions can differ in form]; 'yes, it is scientific notation' — the coefficient constraint was not applied [B1G-M 8.NSO.1.4 misconception]

Question Bank — variation and answers

Variation. Questions 1–4 write in scientific notation with one, two, two and three significant digits (the interior zero in 4 is the trap). 5–6 read notation back. 7 is three real quantities, the third from a calculator display. 8 is the scaffolded comparison in the book's shape, with a scaffold that matches its question. 9 is the guide's two comparisons — one exact, one that must be approximated. 10 compares by inspection, powers first then coefficients. 11 is the diagnostic multiple choice; 12 separates value from form, and 13 asks for the form rule in writing. The Additional sheet mirrors each position.

The bank and its Additional sheet are posted, never assigned; they feed the quizzes. Answers below are the same values the keys print — both come from one source.

Question Bank

- 9×10^3
- 4.5×10^6
- 3.1×10^8
- 7.02×10^9
- 63,000**
- 105,000,000**

7. (a) 1.67×10^{21} molecules (b) 9.3×10^7 miles (c) 400,000,000,000
8. (a) 1.5 (b) 10 (c) $1.5 \times 10 = 15$ times larger
9. (a) 40 times (b) about 89 times
10. (a) 1.2×10^6 — 10^6 is greater than 10^5 , so the powers decide; 1,200,000 > 950,000. (b) 3.8×10^7 — same power, so compare coefficients: $3.8 > 3.09$.
11. A A
12. A, B, C, D A, B, C and D
13. 3.4×10^6 . In scientific notation the coefficient must be at least 1 and less than 10; 34, 340 and 0.34 are not.

Question Bank — Additional

1. 7×10^4
2. 2.8×10^6
3. 6.4×10^8
4. 5.03×10^9
5. 410,000
6. 16,400,000
7. (a) 7.5×10^{18} grains (b) 3.99×10^{13} km (c) 12,000,000
8. (a) 2 (b) 100 (c) $2 \times 100 = 200$ times larger
9. (a) about 4 times (b) about 11.1 times
10. (a) 2.1×10^5 — 10^5 is greater than 10^4 ; 210,000 > 87,000. (b) 5.6×10^6 — same power; $5.6 > 5.09$.
11. A A
12. A, B, C, D A, B, C and D
13. 6.2×10^6 . The coefficient must be at least 1 and less than 10; 62, 620 and 0.62 are not.

What the audit against Math Nation found

- Every item re-derived by sympy at build time (mathcheck: 0 findings); every scientific-notation coefficient scanned (capcheck: 0 findings outside tagged items).
- Math Nation 3.8 student pages verified: 2×10^6 , 2.6×10^7 , 3.01×10^{10} , $6.4 \times 10^7 \div 3.2 \times 10^7 = 2$, $4.2 \times 10^5 \div 2.1 \times 10^4 = 20$, $8.4 \times 10^6 \div 2.1 \times 10^4 = 400$; the Earth-Sun/Moon approximation ($9.3 \times 10^7 \div 2.38855 \times 10^5 \approx 389$) is stated in the book as 'about 4×10^2 '.
- Homework L8 keys: 1.67×10^{21} , $\$1.105 \times 10^8$, 7.5 — all verified. Homework 1a's table ($10^{14} \div 10^{13} = 10$, 10^{24} , 10^{16} , 10^{-1} , 10,000, 100, 999,990,000,000,000,000,000) verified.
- Additional Practice L8 #2 key: $10^6 + 10^4 = 101000$ is wrong (1,010,000). Not reused (audit D2).
- Homework L8 #2 scaffold (a, b) does not lead to its part (c); see Read This First. Rebuilt as bank 8.
- B1G-M items verified: $4 \times 10^5 \div 1 \times 10^4 = 40$; $65,000 \div 730 = 89.04$; $1.44 \times 10^9 \div 1.292 \times 10^8 = 11.15$.

What changed from the book

- 3.8.1 grains-of-sand warm-up → spaced retrieval; the sand number kept in Notes I and Additional 7a.
- 3.8.2-3.8.3 partner explorations (powers of ten, patterns) → Notes I-II from the front.
- 3.8.4 Guided Instruction → Notes II and Example 1; its item 3 (Marcus and the coefficient rule) → whiteboard 9, written.
- 3.8.5 partner collaboration (comparing) → Notes III, Example 2 and whiteboards 6-7.
- 3.8.6 Cool Down ('explain in your own words') → replaced by whiteboard 9, which asks the same thing about a specific case.